Governing the Smart City

From a transparency/ citizen awareness standpoint

- What kinds of public-private partnerships
- Where are the regulatory gaps wrt democratic accountability

Copyright notice

Chennai metro rail

SUCG, STEC 1

Two tunnel boring machines from China were employed to drill from Nehru Park to Egmore and

A team of Chinese workers assembles a Tunnel Boring Machine at the Metro Rail site on Poonamallee High Road near Nehru Park.

Photo: K. Pichumani

Chennai Metro Rail underground tunnelling near Central Station.

The TBMs being used are STEC 1, STEC 2

STEC has 386 patents in China and none in India. No PCTs filed.

<u>Cost</u>: ₹300Cr/km
<u>Total tunnelling cost</u>: ₹11,400Cr.
<u>Average</u>: 14 m/day
<u>Muck</u>: 50 m^3/m

The German company Herrenknecht has deployed eight machines, two machines STEC 1 and STEC 2 are from China and one from the US and one from Japan.

Title: (EN) Tunnel boring machine capable of cutting full section
(ZH) 全断面切削的隧道掘进机

Abstract: (EN) The invention provides a tunnel boring machine capable of cutting a full section, and

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Machine translation

6. (CN102261252) Tunnel boring machine capable of cutting full section

National Biblio. Data | Description | Claims | Drawings | Documents

Permanent Link/ Bookmark:

Application Number: 201110214763.2 **Application Date:** 29.07.2011
Publication Number: 102261252 **Publication Date:** 30.11.2011
Publication Kind : A
IPC: E21D 9/08
Applicants: Shanghai Guanglian Construction and Development Co., Ltd.
上海广联建设发展有限公司
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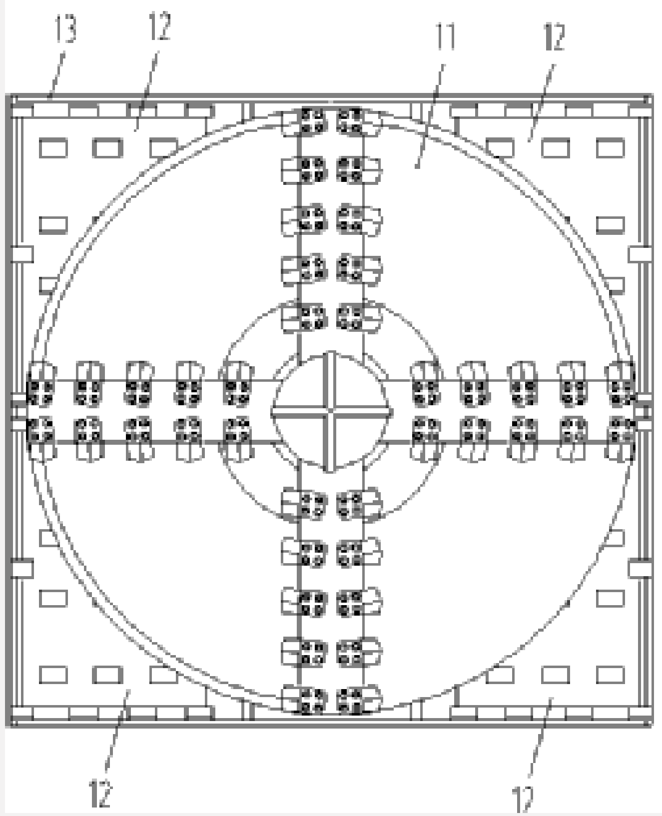
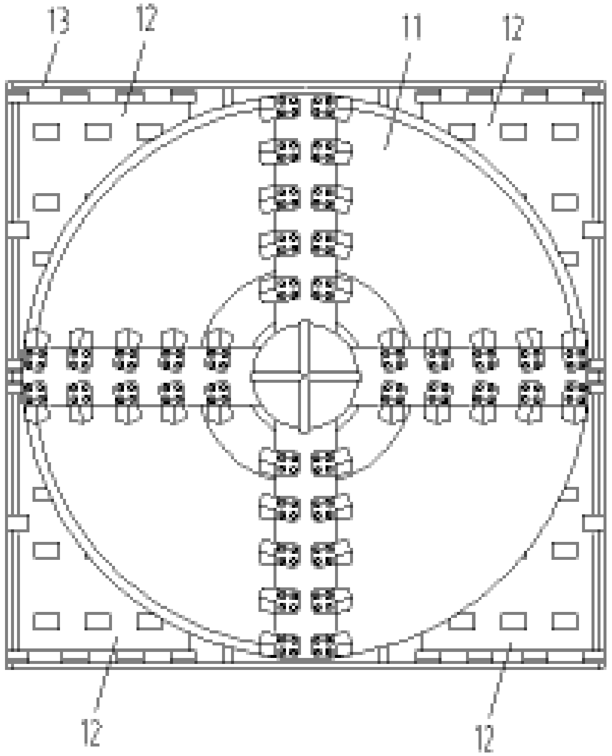
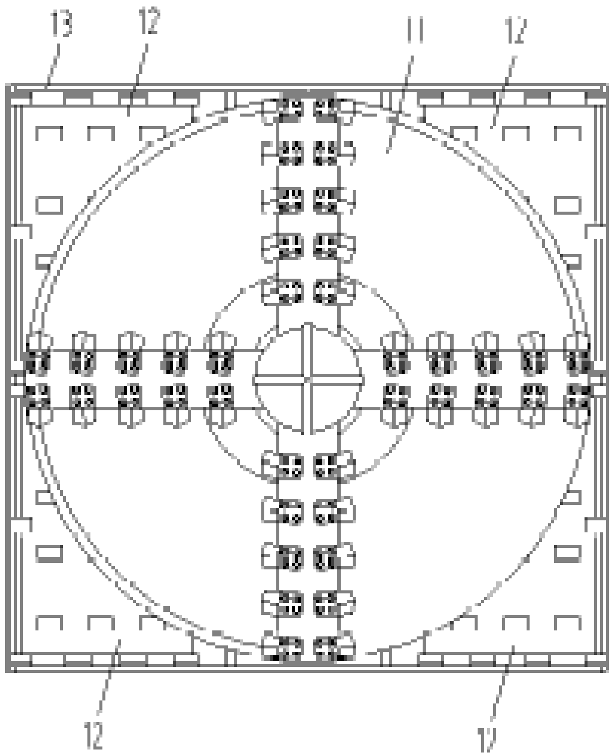
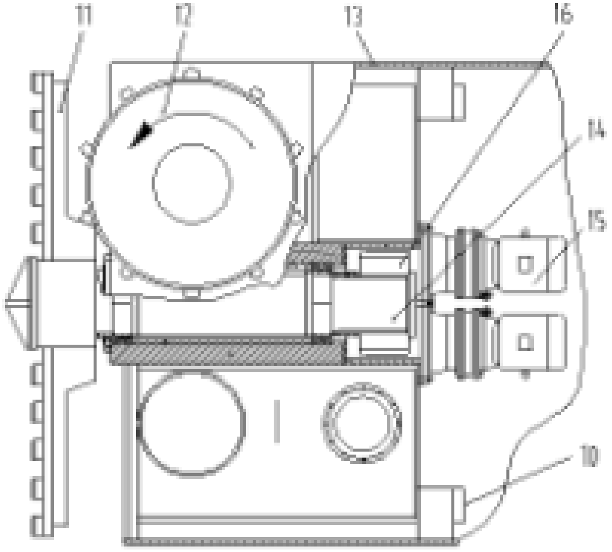
张志勇
Qiao Huashan
乔华山
Wang Zhongbing
王中兵
Shan Yao
闪耀

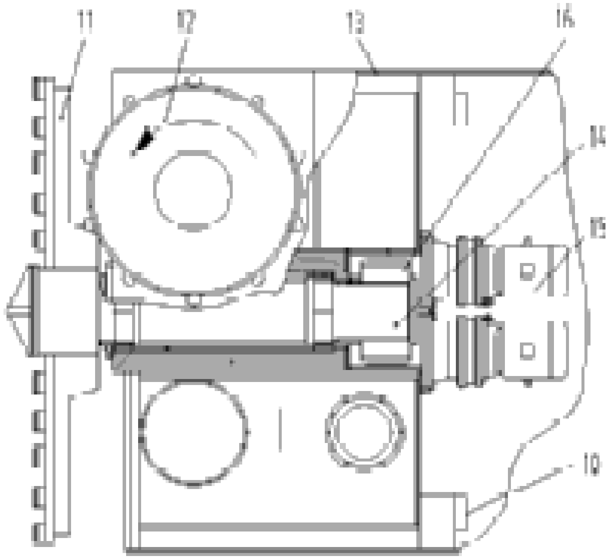


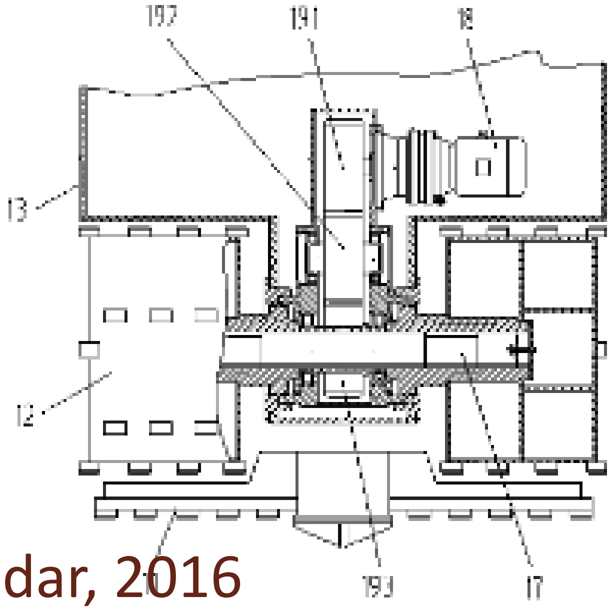
断面切削的技术问题。该掘进机包括壳体和圆盘形的主刀盘，所述主刀盘同轴固定在一主盘转轴的前端，在壳体内设有用于驱动主盘转轴转动的主盘驱动装置，所述壳体上设有至少一根补助盘转轴，各补助盘转轴相互平行，且垂直于主盘转轴，在壳体内设有用于驱动各补助盘转轴转动的多个补助盘驱动装置；每根补助盘转轴至少有一端同轴固定有一个筒状补助刀盘；所述主刀盘的正投影呈圆形，每个补助刀盘的正投影均呈矩形，每个补助刀盘正投影的一角落在主刀盘正投影外，其余部分则落在主刀盘正投影内。本发明提供的掘进机，能提高施工效率，降低施工成本。

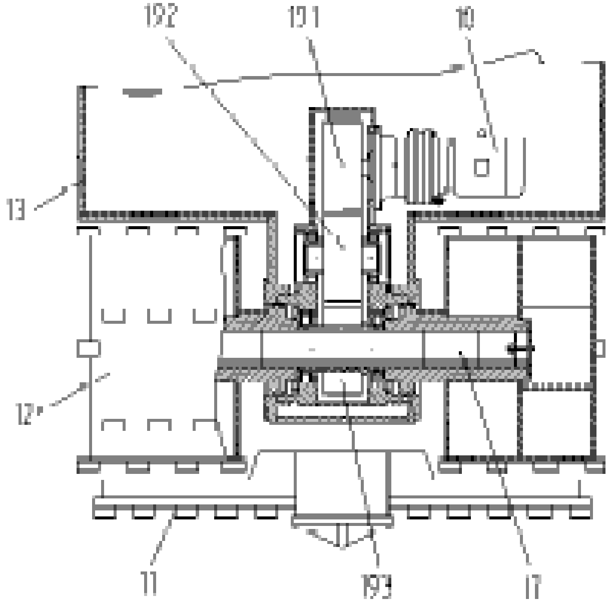












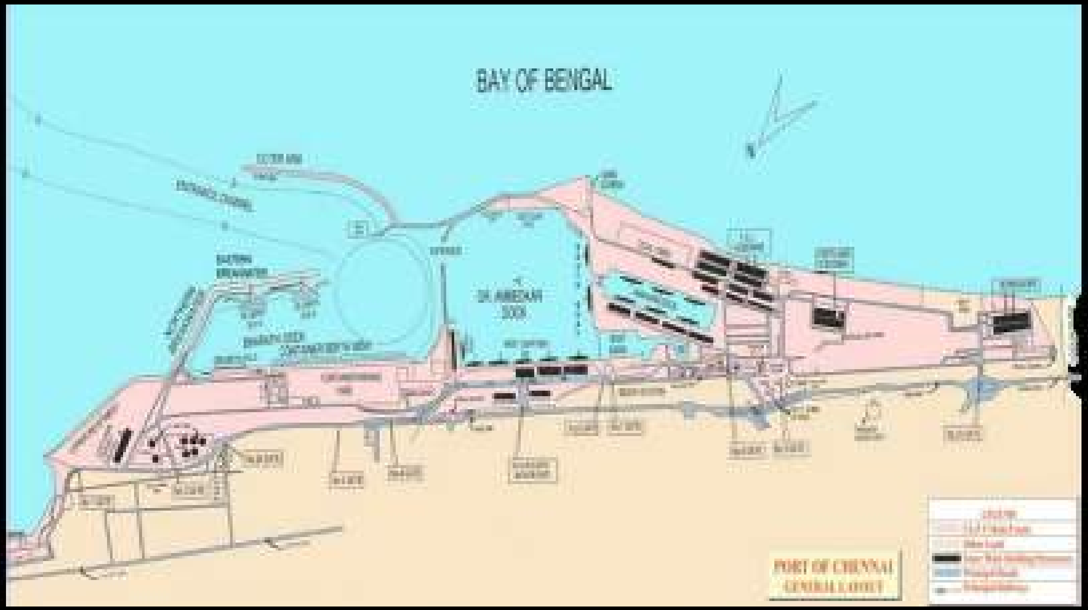

MOSMETROSTROY

1.4% of interest rate and 30 years of repayment period (including 10 years of grace period).

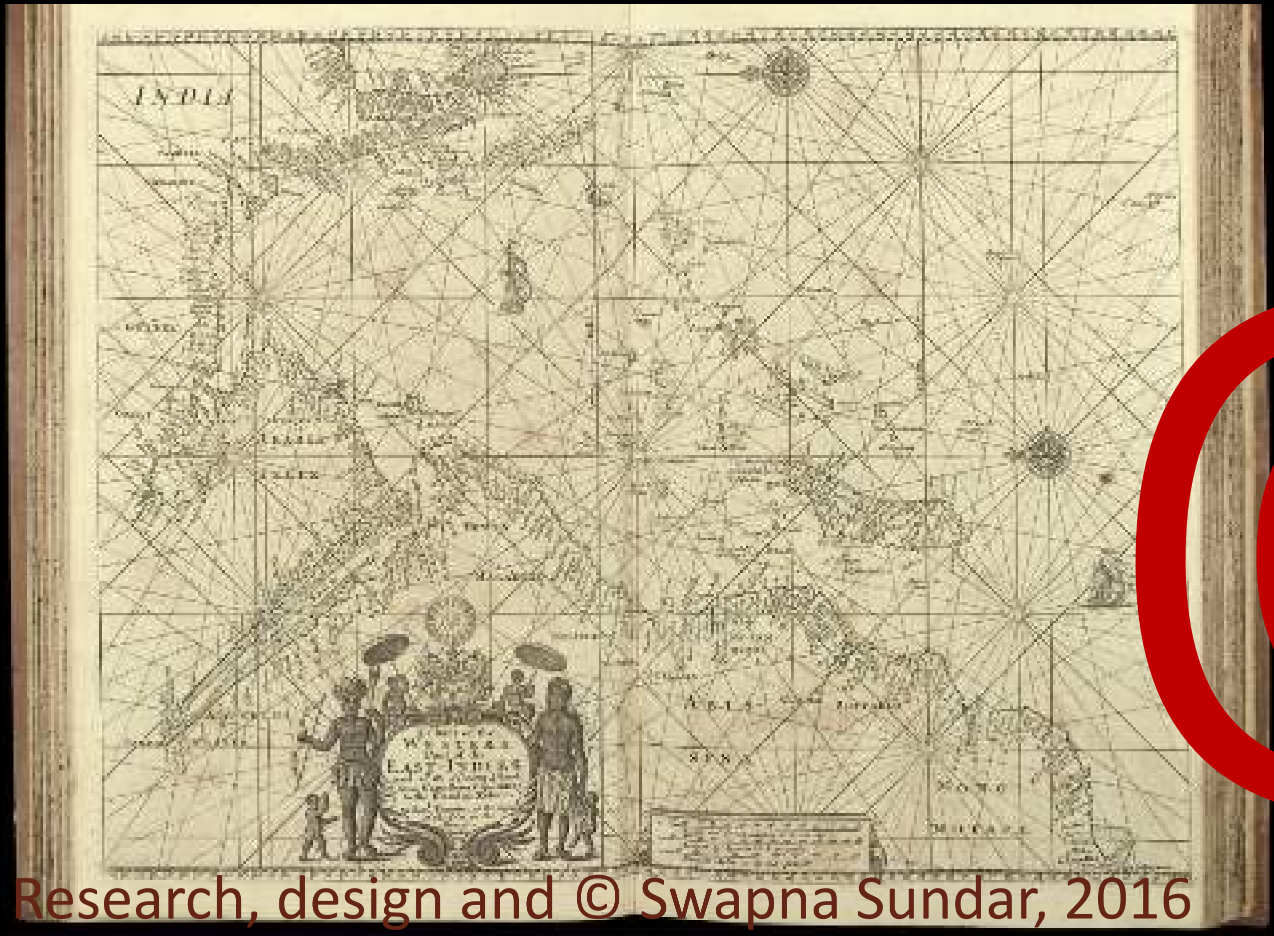
ORV *Sagar Nidhi*

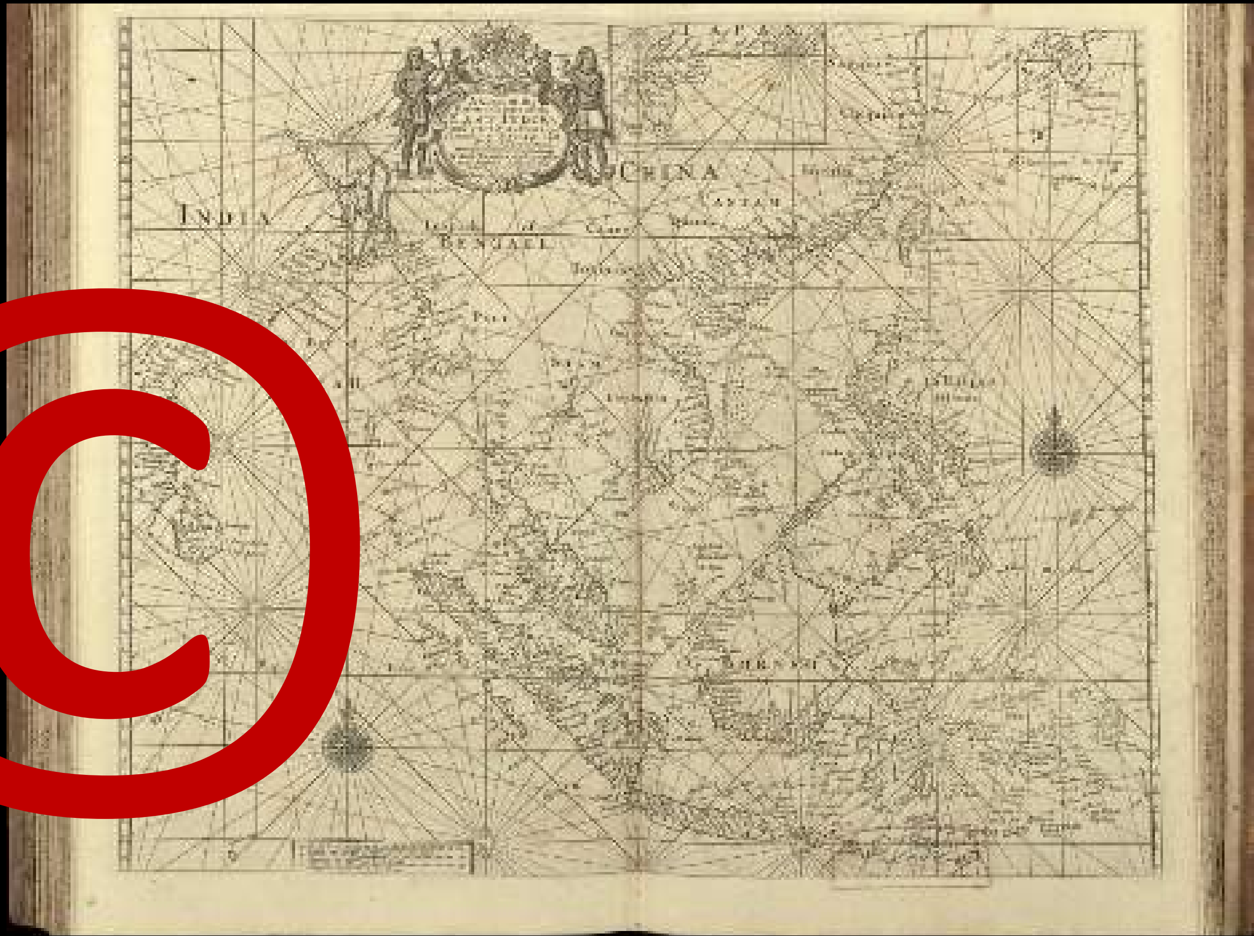
An example of an ENC

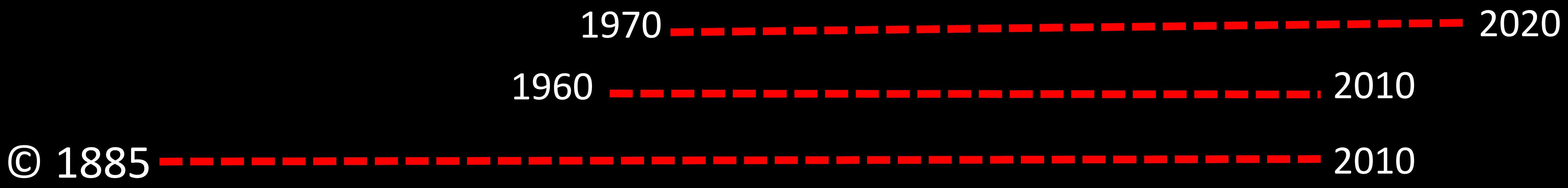
The Harbour of Chennai



The English Pilot, John Thornton, 1703







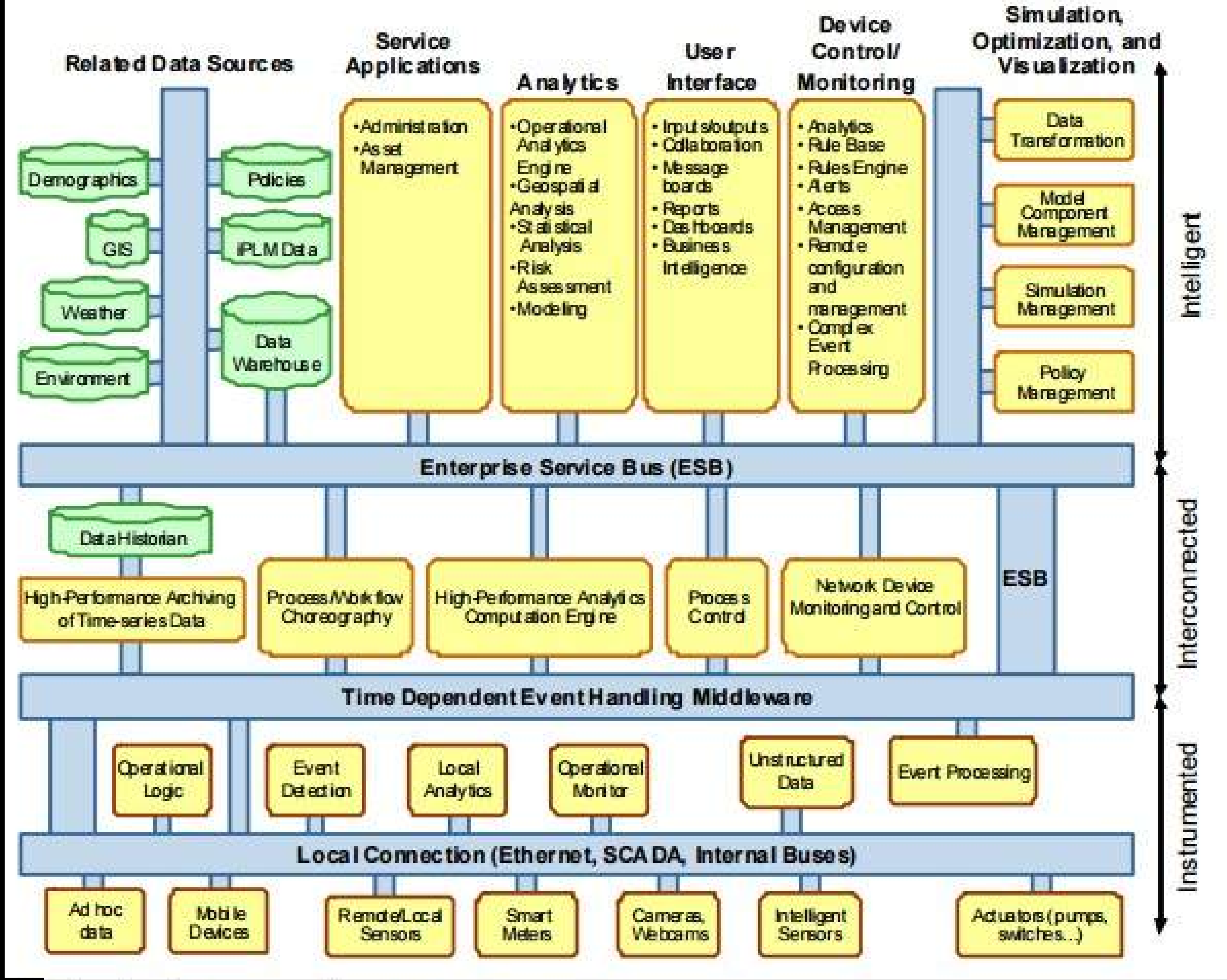
125 years from the end of the calendar year in which it was created

Preserves the status and official version of the documents
Prevents fraud and misrepresentation

…But also ensure financial viability of the official versions

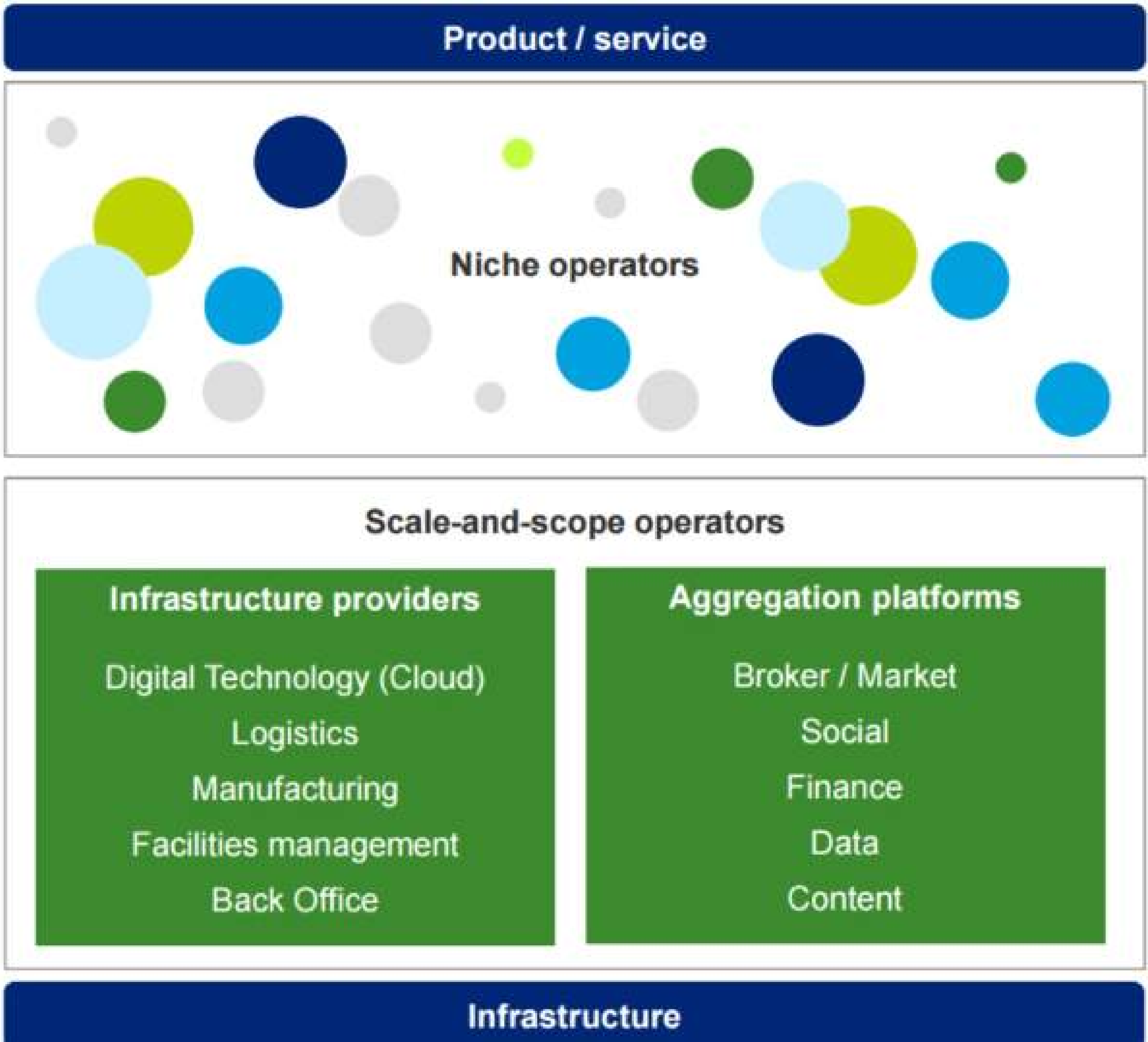

IBM

PwC feels "the pillars of a smart sustainable city" are completely aligned with the proposed UN's Sustainable Development Goals (SDGs).

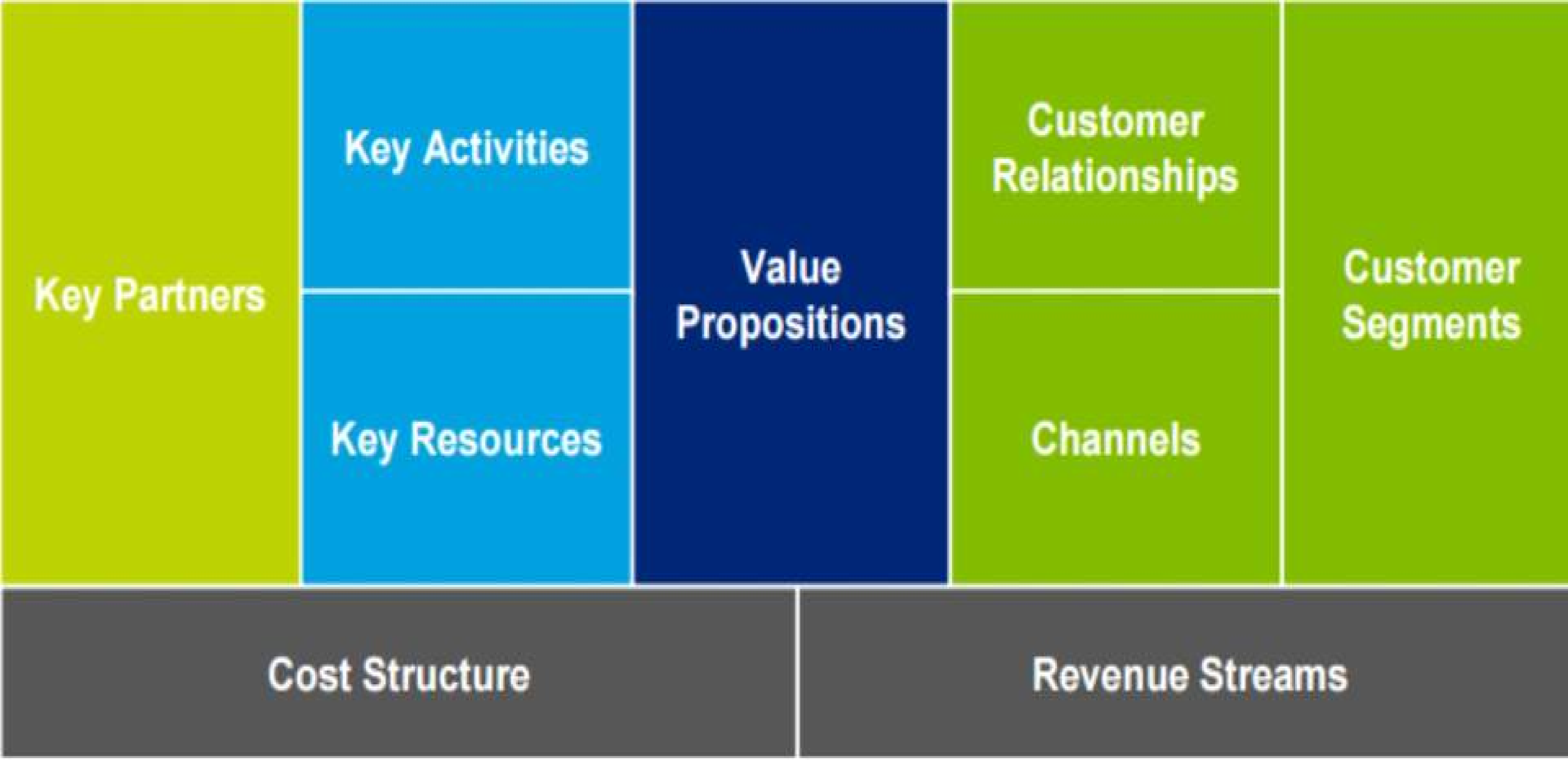


Smarter city component diagram

http://www.redbooks.ibm.com/redpapers/pdfs/redp4733.pdf https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/public-sector/deloitte-nl-ps-smart-cities-report.pdf



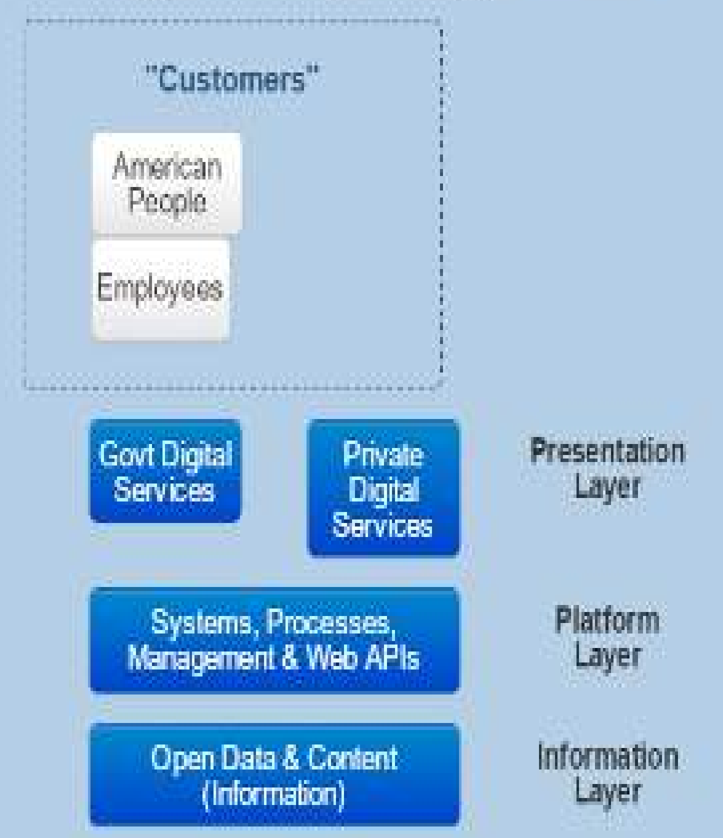
Customers of well-designed services (ref. sl. 2)



Deloitte – Netherlands (ref. sl. 2)

Conceptual Model

Figure 1: The Layers of Digital Services



Before discussing *how* we will build a 21st century digital government, we must first establish a conceptual model that acknowledges the three "layers" of digital services (see Figure 1).

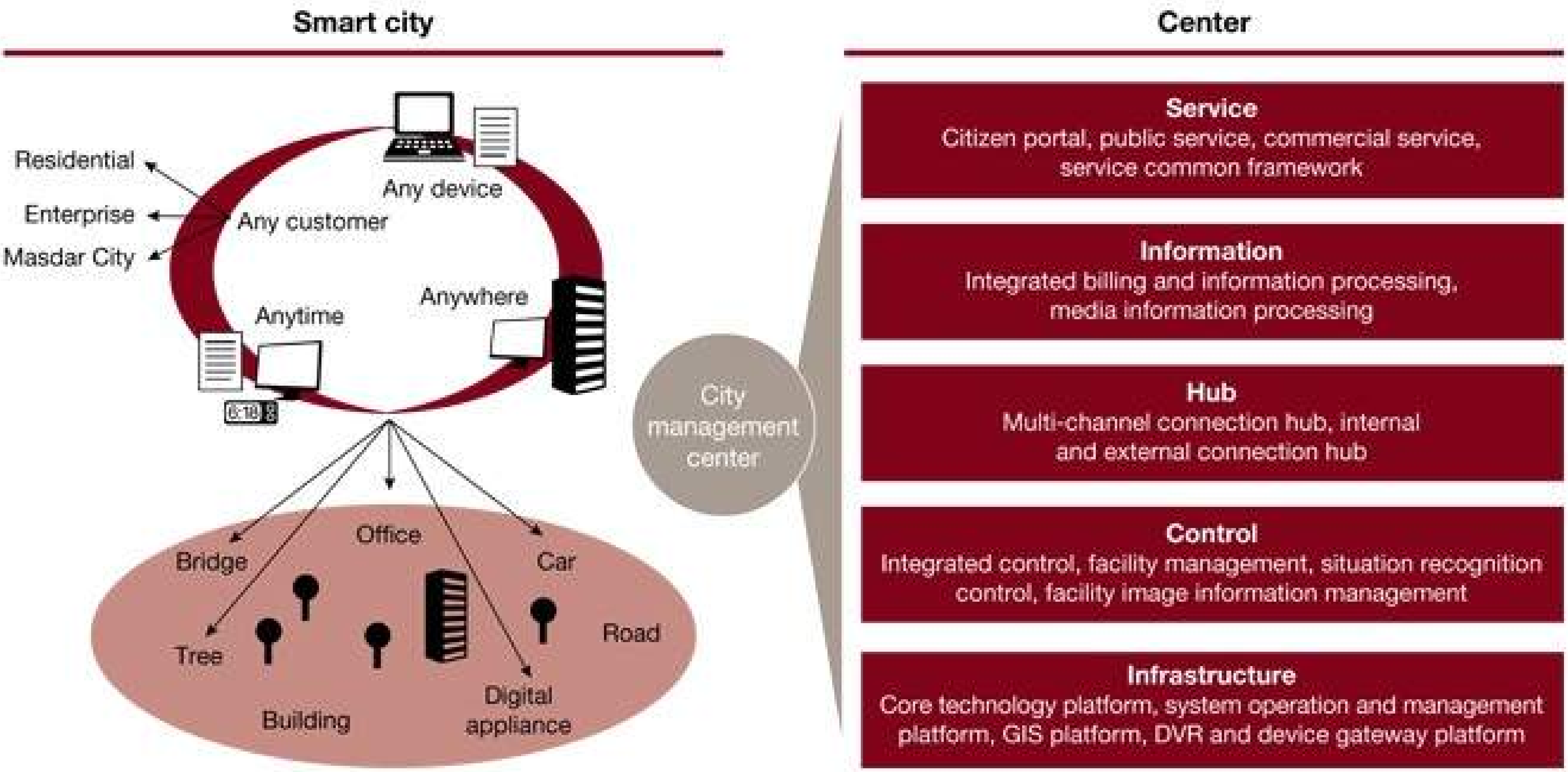
The ***information layer*** contains digital information. It includes structured information (e.g., the most common concept of "data") such as census and employment data, plus unstructured information (e.g., content), such as fact sheets, press releases, and compliance guidance.[14]

The ***platform layer*** includes all the systems and processes used to manage this information. Examples include systems for content management, processes such as web API (Application Programming Interface)[15] and application development, services that support mission critical IT functions such as human resources or financial management, as well as the hardware used to access information (e.g. mobile devices).

The ***presentation layer*** defines the manner in which information is organized and provided to customers. It represents the way the government and private sector deliver government information (e.g., data or content) digitally, whether through websites,[16] mobile applications, or other modes of delivery.

These three layers separate information *creation* from information *presentation* – allowing us to create content and data once, and then use it in different ways. In effect, this model represents a fundamental shift from the way our government provides digital services today.

https://www.whitehouse.gov/sites/default/files/omb/egov/digital-government/digital-government.html



Source: Strategy&

May 13, 2015: Commerce and Industry Minister Nirmala Sitharaman informed the Parliament that JICA has been entrusted the task of preparation of comprehensive regional perspective plan, including master plan and development plan for selected industrial notds, along the Chennai-Bengaluru Industrial Corridor, as the nodal agency of the CBIC project. (http://articles.economictimes.indiatimes.com/2015-05-13/news/62124553_1_delhi-mumbai-industrial-corridor-jica-draft-master)

MAKE IN INDIA: PM says cities were built on riverbanks, today along highways, and in the future they will e built based on availability of optical fiber networks and next-generation infrastructure. India's Smart City plan is part of a larger agenda of creating Industrial Corridors between India's big metropolitan cities in India. These include the Delhi-Mumbai Industrial Corridor, the Chennai-Bangalore Industrial Corridor and the Bangalore-Mumbai Economic Corridor. It is hoped that many industrial and commercial centres will be recreated as "Smart Cities" along these corridors. (http://www.makeinindia.com/article/-/v/internet-of-things)

Industrial Corridors under Development (DMIC & CBIC)

Delhi-Mumbai Industrial Corridor

Amritsar Kolkata Industrial Corridor

Bengaluru-Mumbai Economic Corridor

Chennai-Bengaluru Industrial Corridor

DMIC and CBIC are being supported by GoI and GoJ.

National Manufacturing Plan Targets

- *~15% y-o-y* growth in manufacturing sector to achieve *25%* contribution to GDP by 2022
- *100 million* jobs by 2022
- *Skill development* for inclusive growth
- Improved *technology* orientation & *value addition*
- Global Competitiveness
- Environmental sustainability

Challenges for Investment (1) – Doing Business

Behind ASEAN and other emerging coutries

Doing Business 2015, World Bank

	Rank
Thailand	26
Russia	62
China	90
Vietnam	78
Brazil	120
India	**142**
Indonesia	144
Bangladesh	173

Challenges for Investment (2) – Global Competitiveness

Global Competitiveness Index

	Rank (out of 140)	Score (1–7)
GCI 2015–2016	**55**	**4.3**
GCI 2014–2015 (out of 144)	71	4.2
GCI 2013–2014 (out of 148)	60	4.3
GCI 2012–2013 (out of 144)	59	4.3
Basic requirements (60.0%)	**80**	**4.4**
1st pillar: Institutions	60	4.1
2nd pillar: Infrastructure	81	3.7
3rd pillar: Macroeconomic environment	91	4.4
4th pillar: Health and primary education	84	5.5
Efficiency enhancers (35.0%)	**58**	**4.2**
5th pillar: Higher education and training	90	3.9
6th pillar: Goods market efficiency	91	4.2
7th pillar: Labor market efficiency	103	3.9
8th pillar: Financial market development	53	4.1
9th pillar: Technological readiness	120	2.7
10th pillar: Market size	3	6.4
Innovation and sophistication factors (5.0%)	**46**	**3.9**
11th pillar: Business sophistication	52	4.2
12th pillar: Innovation	42	3.6

Stage of development

Source: Global Competitiveness Report 2015-16, World Economic Forum

JICA's Cooperation Strategy for India

India's Development Strategy

"Faster Growth"

Employment Generation

"Inclusive Growth"

Global Environmental Issues

JICA's Cooperation Strategy for India

Stable Energy Supply
- Improvement of Energy Supply Capacity,

Improvement of Transportation Networks
- Strengthening Major Transport Networks (Railways/Roads/Airports/Ports)
- City Transport System (Metro/Outer Ring Roads)

Employment Generation
- Private Sector Assistance

Income Generation in Rural Areas
- Improvement of Agricultural Productivity

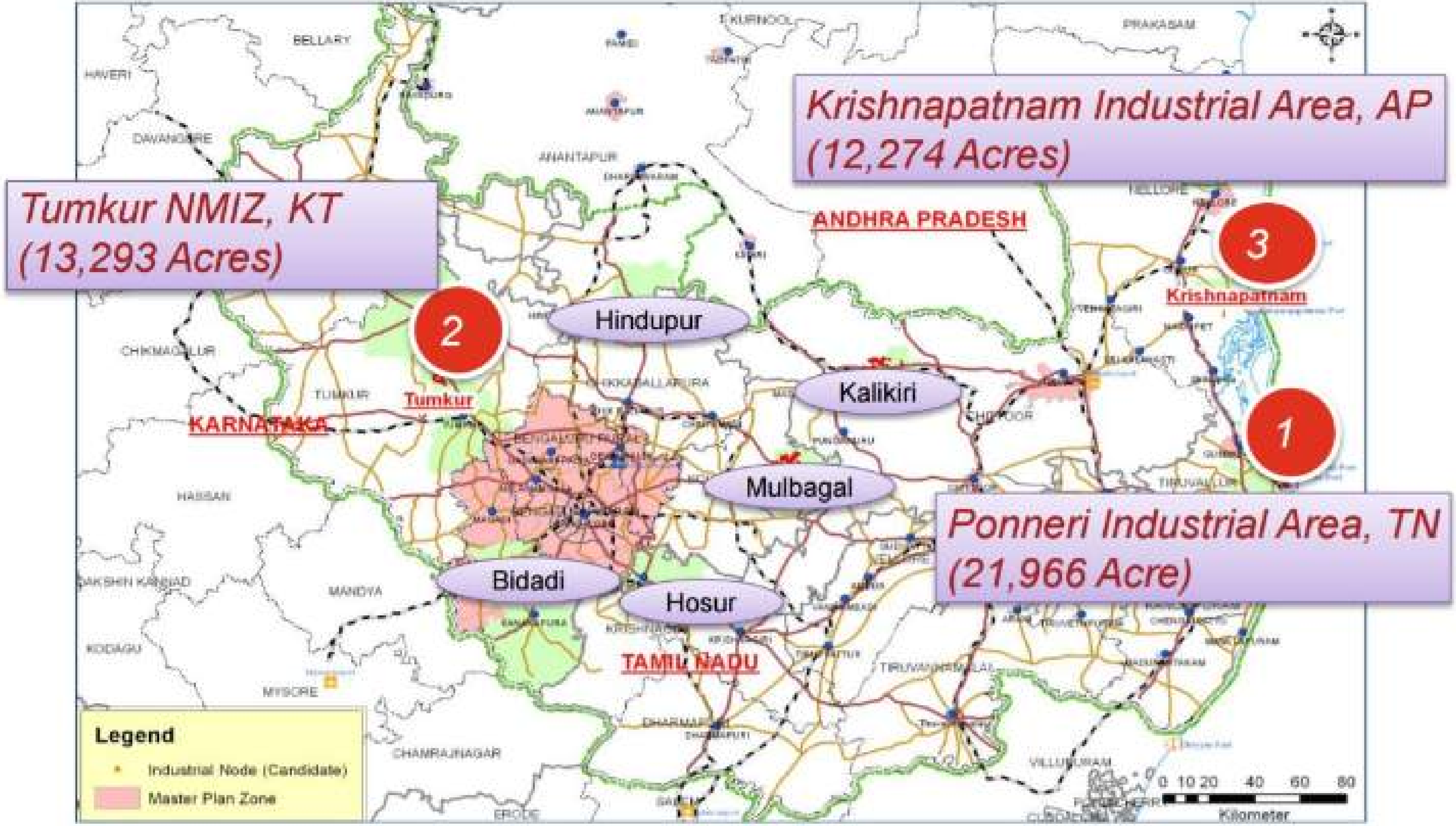
Improvement of Basic Social Services
- Improvement of Basic Social Services (Health and Sanitation)

Conservation and Improvement of Urban Environment
- Pollution Prevention and Management
- Water Resources Management

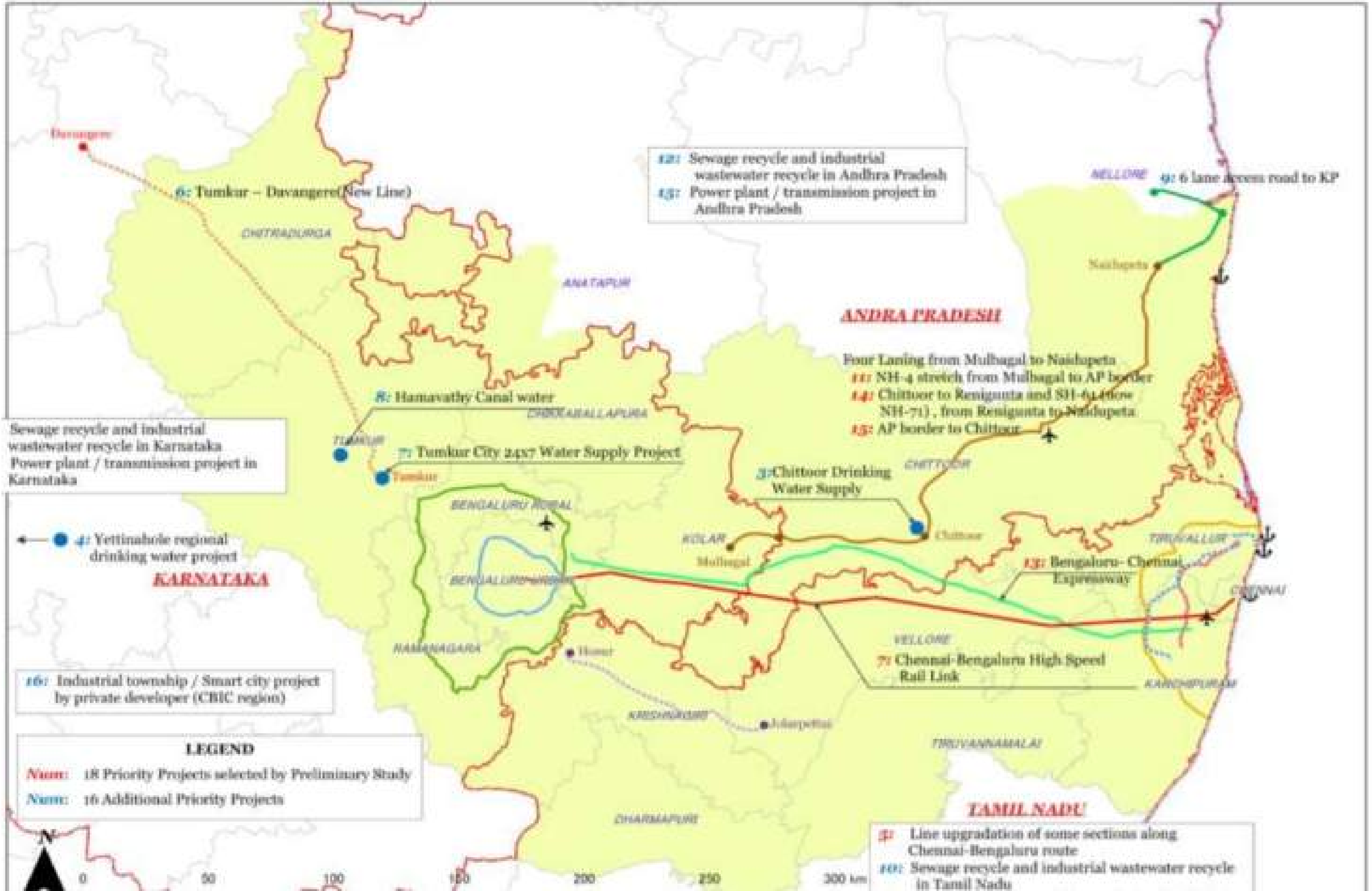
Environmental Conservation
- Afforestation, Forest Preservation, Bio-diversity Conservation
- Energy Efficiency and Conservation
- Renewable Energy

Chennai-Bengaluru Industrial Corridor

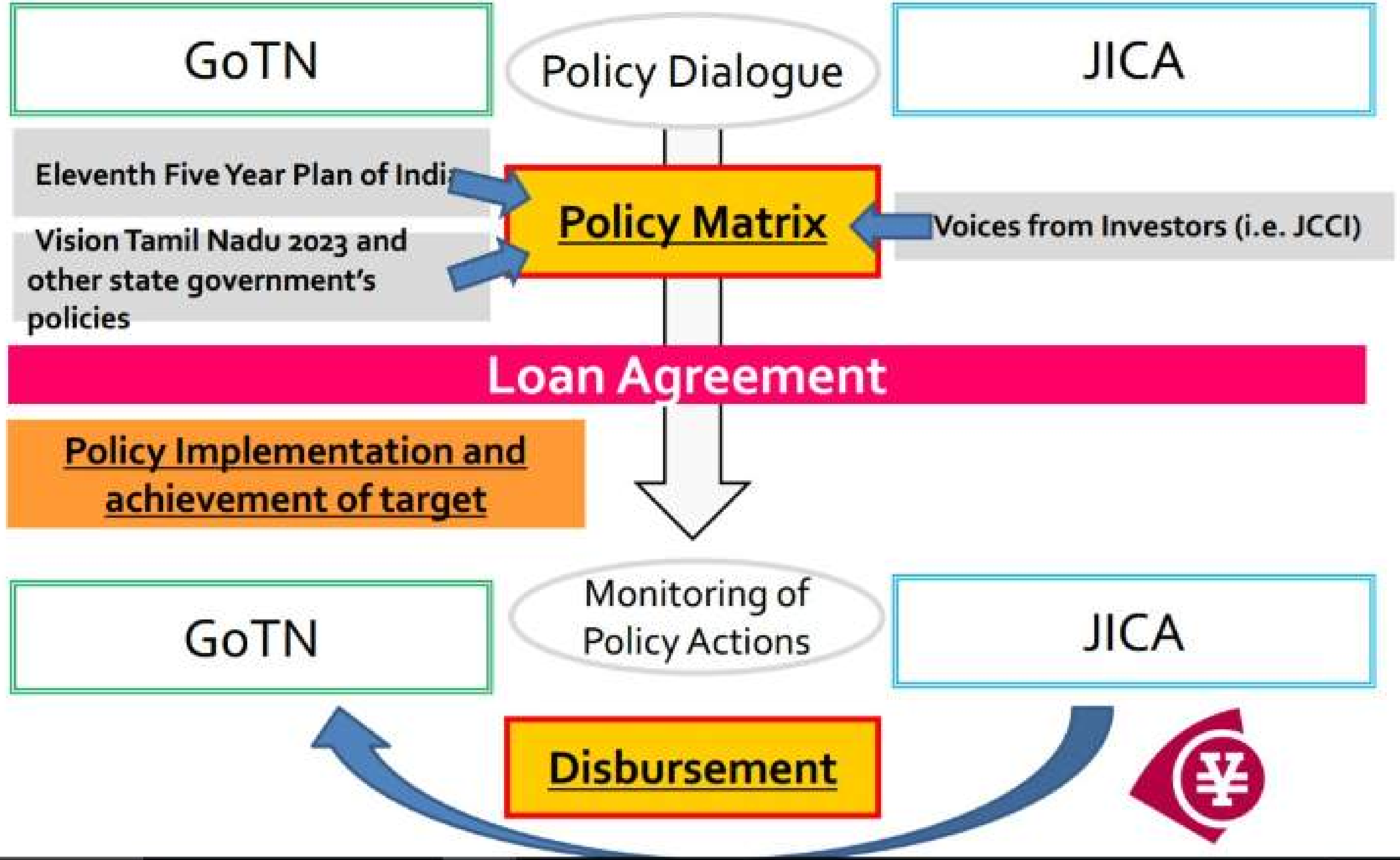
<u>Eight Nodes</u> have been identified based on land availability and growth potential.
<u>Three Nodes</u> have been selected for prioritised implementation

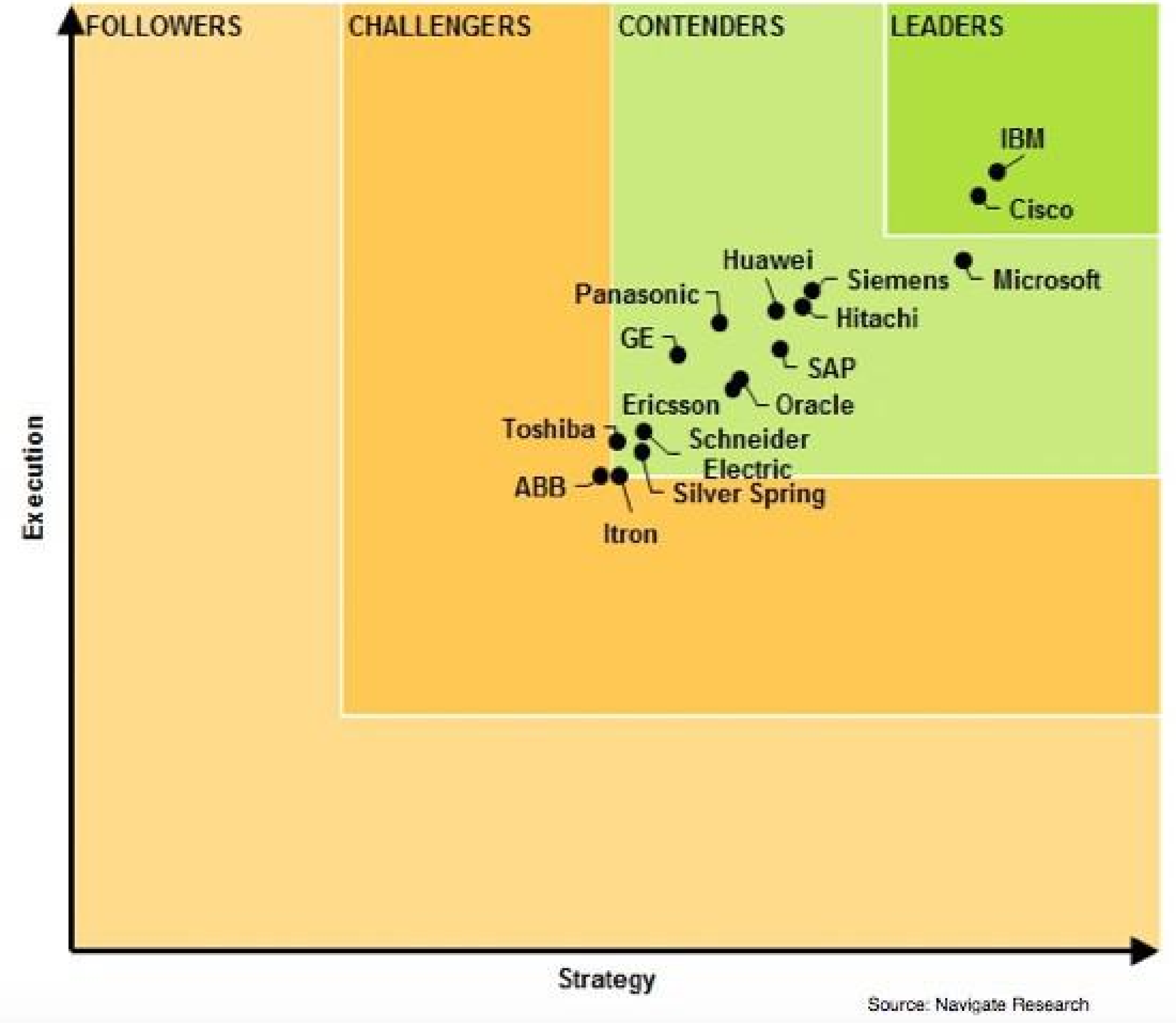


Regional M/P : Priority Infrastructure (Road, Water Supply, Power, etc)



Basic Concept of Program Loan





IBM
Cisco
Microsoft
Siemens
Hitachi
Huawei
SAP
Panasonic
Oracle
Ericsson

Read more: http://www.environmentalleader.com/2016/03/02/top-smart-city-technology-suppliers-announced/#ixzz4PyX54QT7

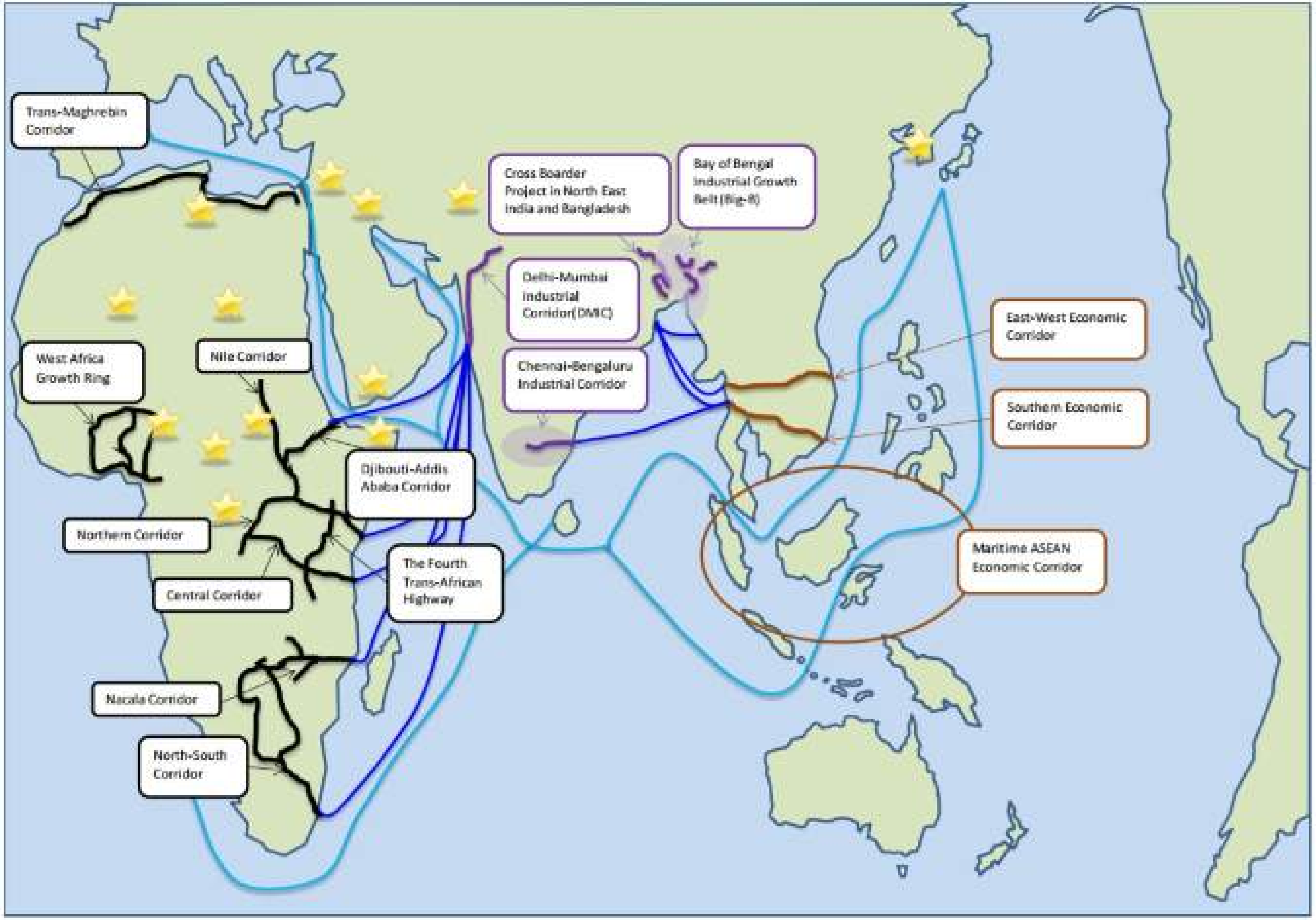
Smart Cities Council India:

- IBM
- ZTE
- AT&T
- Sensus
- Cisco
- S&C Electric
- Dow Building and Construction
- Microsoft
- CH2M
- Schneider
- Qualcomm
- SAS
- Oracle
- Daimler
- Deloitte
- Itron
- Allied Telesis
- GE

http://india.smartcitiescouncil.com/

Available in other flavours also…

JICA's activities for prosperity of the Indo-Pacific



- widening internet access for residents (from reducing digital divides to increasing connectivity options in public places)
- creating channels for more open government (improving council transparency and using visualisation tools to make data more visible to the public)
- citizen engagement (an improved city website, and the use of social networking tools to improve citizen dialogue, including crowdsourcing problems such as potholes)
- and industry support (a mix of seed grant funding for start-ups, tax incentives
- and state-sanctioned real estate developments aimed at meeting the locational needs of technology and new media companies).

Fig. 2: A satirical cartoon depicting representatives of companies and organisations involved in construction. They are pushing a fisherman off a ship labelled 'Tidal Power Plant Development', saying: 'All that needs to happen is you taking the plunge!'

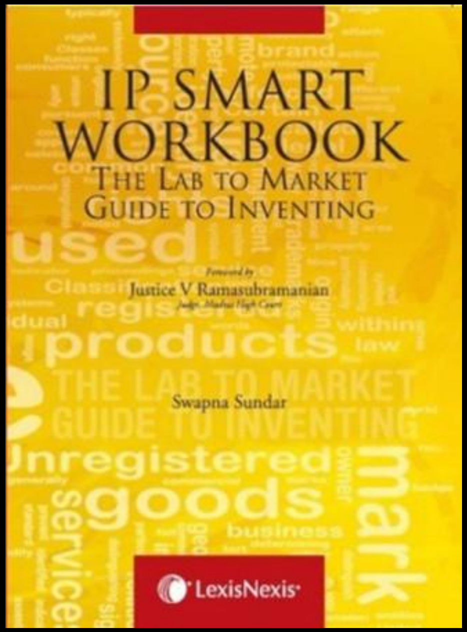
Fig. 3: A satirical cartoon depicting a teacher and her students in a museum. The teacher explains to her students which creatures used to live on the territory that now houses Songdo. A fisherman figures among the extinct marine creatures.

- to implement better living conditions in large metropolises
- to involve citizens in city government
- to support sustainable economic development and city attractiveness
- to build a long-term smart strategy capable of producing better returns from investments and deciding priorities regarding each city
- The survey shows that no consolidated standards or best practices for governing SCs
- Choice of residents in Songdo… (http://songdoibd.com/live/)

Thank you for your attention

For more information, research, reference on aspects of

- IPR
- Technology integration
- Legal/constitutional impact of these projects



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